Problem Sheet 9

1.

What thermal noise voltage do you expect from a 1 MΩ resistor measured at room temperature with an instrument which has a 1 kHz bandwidth?

2. SEMINAR PROBLEM I

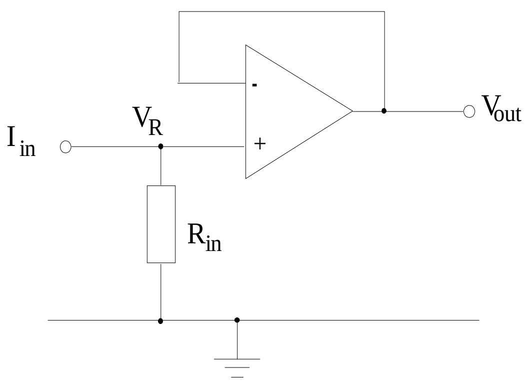
(a) If 100 V is applied to a forward-biased diode in series with a 10 Ω resistor, how much shot noise is generated when measured with an instrument of bandwidth 100 kHz?

(b) If the shot noise signal in (a) is taken across the resistor (i.e. as a voltage) and amplified with an amplifier of gain 10^5, what will be the voltage noise at the amplifier output? (Assume that the amplifier is 'ideal' and AC-coupled to the resistor)

(c) At room temperature, what will be the thermal noise at the output of the amplifier?

3.

(a) The current from a vacuum tube sensor (e.g. a PMT) is converted to a voltage across a resistor and buffered with a voltage follower. The system has minimal flicker noise, and the amplifier used has a noise factor $F = 1$. What does this value of noise factor mean for the performance of the amplifier, and is it physically achievable in a real device?



(b) Show that the RMS noise voltage at V_{out} is dominated by shot rather than thermal noise for a voltage V_R across R_{in} greater than $2k_BT/e$.

(k_B is Boltzmann's constant, T the temperature in K, and e the electronic charge).

4. SEMINAR PROBLEM II

(a) A 10 kΩ resistor R_{in} forms the input impedance of an instrument that consists of an input stage with an amplifier of noise factor $F = 4$. The gain of the amplifier stage is a near constant from 100 Hz to 100 kHz, and falls abruptly beyond these points. Assuming thermal noise dominates, what current must flow through the resistor to give a signal to noise ratio of 20dB at the output with R_{in} initially at 25°C?

(b) To what temperature must R_{in} be cooled to in in order to improve the SNR to 40 dB?

5.

(a) In flicker noise, what value does the noise power per Hz approach as the frequency f of a measurement approach zero? Why as a result of this are we ***not*** immediately swamped by flicker noise every time we carry out a DC measurement?

(b) Two identical DC coupled amplifiers with 10 kHz upper frequency limits have been on for 1 and 100 days, respectively. What is the ratio of flicker noise power in the output of these two units?

Numerical Answers

1 about 4μV RMS

2 (a) 0.56μA RMS (b) $0.56V_{RMS}$ (c) $0.01V_{RMS}$

4 (a) $8.2nA_{RMS}$ (b) 3K

5 (b) 1.22